MATH 1075 - Midterm 2 Review Problem

10/13/2026

1 Section 7.3-7.5

1. $\frac{x}{4x+2} + \frac{3x}{6x+3}$

2. $\frac{x}{x^2-9} - \frac{4}{x-3}$

3. $\frac{\frac{12x^2}{5y}}{\frac{8x^6}{9y^2}}$

4. $\frac{1-\frac{4}{x^2}}{1-\frac{2}{x}-\frac{8}{x^2}}$

2 7.6-7.7

1. $\frac{7}{4x} = \frac{3}{x-5}$

2. $\frac{2}{x+3} = \frac{5}{4x+12} - \frac{3}{8}$

3. According to the website for the state of Virginia, 0.8 million tons of clothing is reused or recycled out of 7 million tons of clothing discarded. If 17.5 million tons of clothing is discarded, how many tons will be reused or recycled?

4. A fishing boat travels 9 mi downstream with the current in the same amount of time that it travels 3 mi upstream against the current. If the speed of the current is 6 mph, what is the speed at which the boat travels in still water?

5. Floyd can walk 2 mph faster than his wife, Rachel. It takes Rachel 3 hr longer than Floyd to hike a 12-mi trail through a park. Find their speeds.

6. A pump can empty a small pond in 4 hr. Another more efficient pump can do the job in 3 hr. How long would it take to empty the pond if both pumps were used?

3 Section 2.8

Solve the inequality and plot it on number line. Write your answer in interval notation and in set notation.

1. $-2x + 5 \geq -x + 5$

2. $-6 \leq 4x - 2 < 12$

4 Section 8.1-8.2

Find domain of each function.

1. $f(x) = \frac{(x+1)^2}{(x+1)}$

2. $g(x) = \frac{5x-1}{x^2+1}$

3. $h(x) = \sqrt{x+5}$

4. $k(x) = \frac{1}{\sqrt{x+5}}$

5. $f(x) = x^2 + 5x + 6$

Given $f(x) = x^2 + x + 1$, evaluate:

1. $f(0)$

2. $f(1)$

3. $f\left(-\frac{1}{2}\right)$

4. $f(-2)$

5 Section 8.3-8.4

Given $f(x) = \sqrt{x+4}$ and $g(x) = x^3$, find:

1. $(f+g)(5)$

2. $(f \cdot g)(2)$

3. $\left(\frac{f}{g}\right)(12)$

4. $(f \circ g)(2)$

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5. $(g \circ g)(2)$

6. $(f \circ g)(x)$

7. $(g \circ f)(x)$

8. $(g \circ g)(x)$

9. $(f \circ f)(x)$